

Elimination with Matrices

Solve $Ax = b$ the way every computer does: eliminate down to an upper triangular U , then back-substitute. And see every step as a matrix.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. The idea of elimination

Turn the system into a triangle you can solve from the bottom up. Pivots and multipliers.

2. 2×2 by elimination

The two-fund portfolio in one step, and what elimination does to the row picture.

3. 3×3 , step by step

The three-fund portfolio: augmented matrix, three elimination steps, the upper triangular U .

4. Back substitution

Solve $Ux = c$ from the last equation up.

5. When elimination breaks down

A 0 in the pivot position: fixed by a row exchange, or a sign the matrix is singular.

6. Elimination as matrices

Each step is a multiplication by an elimination matrix E . Row times matrix combines rows.

7. Row exchanges: permutations

The matrix P that swaps rows, and why $PA \neq AP$.

8. Undoing a step: inverses

Every step can be reversed, which is why elimination gives the right answer.

PART 2 Prerequisites

From Lesson 1: The Geometry of Linear Equations. Click any link to review it, then come back.

- Matrix form $Ax = b$: rows of A are rules, columns are funds.
- Matrix $\times$ vector, two ways: Ax is a combination of the columns of A .
- The row picture: each equation is a line (or plane).
- The three-fund portfolio (3×3): the system we solve here.
- Singular matrices: columns that add no new direction.

THE RUNNING EXAMPLE: THE SAME PORTFOLIO, SOLVED PROPERLY

In Lesson 1 we split ₹10 lakh between funds to hit a budget, a return and a market exposure. We found the answers by drawing pictures and by spotting tricks. That doesn't scale: with 50 funds and 50 rules nobody can draw or guess.

This lesson gives the method every computer uses: **elimination**. We apply it to the same two systems:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}}_{\text{Nifty + bonds}} = \begin{bmatrix} 10 \\ 105 \end{bmatrix} \quad \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{Nifty + bonds + midcap}} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}$$

Rows: budget (₹ lakh), return (₹ thousand), market exposure (₹ lakh). Unknowns: ₹ lakh in each fund.

1. The idea of elimination

Some systems are easy. If the last equation has only one unknown, you solve it at once, then work upwards:

$$\begin{aligned} x + y + z &= 10 \\ -5y + 3z &= -9 \\ -0.4z &= -0.8 \end{aligned} \quad \Rightarrow \quad z = 2, y = 3, x = 5.$$

Its matrix is upper triangular: zeros everywhere below the diagonal. **Elimination** turns any (good) system into one like this, without changing its solution.

ELIMINATION IN ONE SENTENCE

Use the first equation to remove x from every equation below it; then use the new second equation to remove y from every equation below it; and so on, until the matrix is upper triangular. Strang: "it's the way every software package solves equations."

Two words do all the work:

- The pivot is the diagonal entry we use to clear the entries below it.

- The multiplier is how many times we subtract the pivot row: **multiplier** = (entry to clear) $\div$ (pivot). The multiplier for row i , column j is written ℓ_{ij} ("ell").

WHY SUBTRACTING EQUATIONS IS ALLOWED

If two statements are true, their difference is true too. Subtracting $12 \times$ (budget rule) from the return rule gives a new rule that every correct portfolio still satisfies, and we can undo it by adding back (topic 8). So the solution never changes.

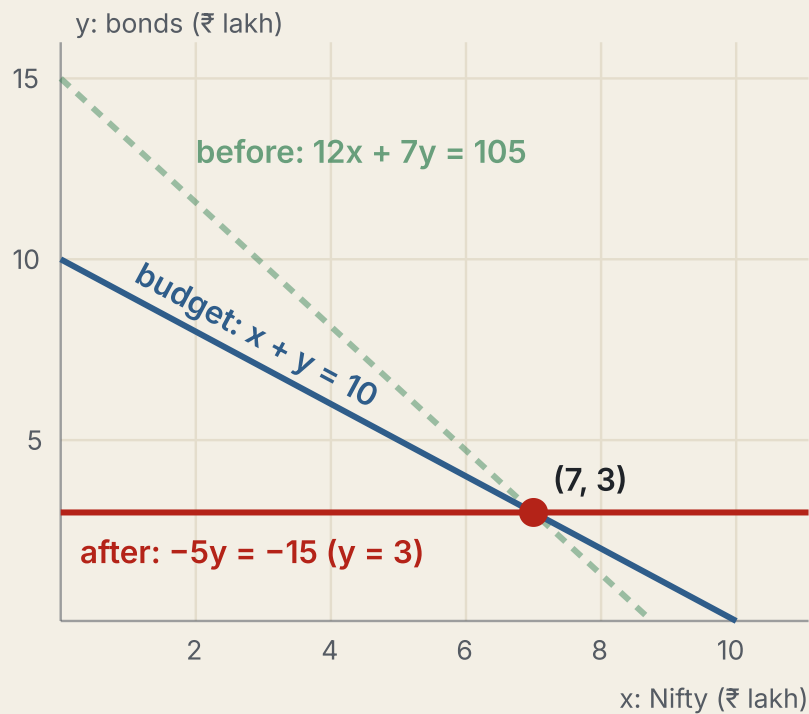
2. 2×2 by elimination

The two-fund portfolio: $x + y = 10$ (budget) and $12x + 7y = 105$ (return).

- 1 **Pivot:** the 1 in the top-left corner.
- 2 **Multiplier:** the entry to clear is 12, so $\ell_{21} = 12/1 = 12$.
- 3 **Subtract** $12 \times$ row 1 from row 2, on both sides: $(12x + 7y) - 12(x + y) = 105 - 12(10)$, which is $-5y = -15$.

$$\left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 12 & 7 & 105 \end{array} \right] \xrightarrow{R_2 - 12R_1} \left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 0 & \boxed{-5} & -15 \end{array} \right]$$

Now back-substitute: $-5y = -15$ gives $y = 3$, then $x + 3 = 10$ gives $x = 7$. **₹7 lakh Nifty, ₹3 lakh bonds**, the same answer as Lesson 1. The pivots are 1 and -5 .



What elimination does to the row picture. The return line (dashed) is replaced by the horizontal line $y = 3$. Different line, **same crossing point** $(7, 3)$: the solution didn't change, it just became easy to read.

▼ **Check yourself (MIT problem 2.1):** eliminate $2x + 3y = 5$, $6x + 15y = 12$. What are the multiplier and the pivots?

$\ell_{21} = 6/2 = 3$. Row 2 $- 3 \times$ row 1 gives $6y = -3$. Pivots 2 and 6. Back-substitute: $y = -\frac{1}{2}$, then $2x = 5 + \frac{3}{2}$, so $x = \frac{13}{4}$.

3. 3×3 , step by step

Write the coefficients and the right-hand side side by side. Tacking b on as an extra column gives the augmented matrix $[A \mid b]$, so whatever we do to an equation's left side we also do to its right side.

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array} \right] \quad \begin{array}{l} \text{budget} \\ \text{return} \\ \text{exposure} \end{array}$$

Column 1: clear everything below the first pivot

The first pivot is the 1 in the top-left corner.

- **Step (2, 1):** $\ell_{21} = 12/1 = 12$. Row 2 $- 12 \times$ row 1: $(12, 7, 15, 111) - (12, 12, 12, 120) = (0, -5, 3, -9)$.
- **Step (3, 1):** $\ell_{31} = 1/1 = 1$. Row 3 $- 1 \times$ row 1: $(1, 0, 1.2, 7.4) - (1, 1, 1, 10) = (0, -1, 0.2, -2.6)$.

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & -5 & 3 & -9 \\ 0 & -1 & 0.2 & -2.6 \end{array} \right]$$

The Nifty amount x is now gone from rules 2 and 3. They form a smaller 2×2 system in y and z , and we simply repeat.

Column 2: clear below the second pivot

The second pivot is the -5 that has appeared in row 2, column 2.

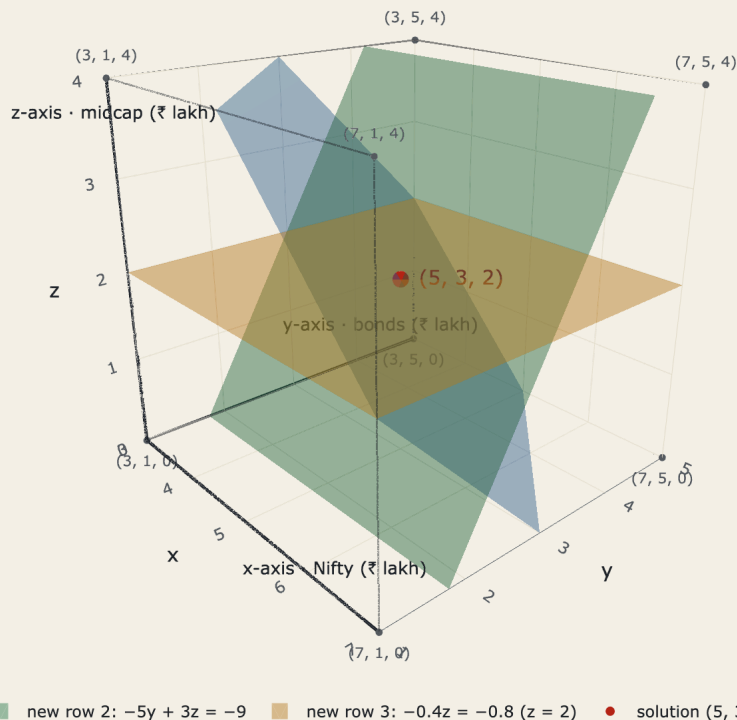
- **Step (3, 2):** $\ell_{32} = (-1)/(-5) = 0.2$. Row 3 $- 0.2 \times$ row 2: $(0, -1, 0.2, -2.6) - (0, -1, 0.6, -1.8) = (0, 0, -0.4, -0.8)$.

$$[U \mid c] = \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & 3 & -9 \\ 0 & 0 & \boxed{-0.4} & -0.8 \end{array} \right]$$

RESULT OF ELIMINATION

A has become the upper triangular matrix U , and b has become $c = (10, -9, -0.8)$. The new system $Ux = c$ has **exactly the same solution** as $Ax = b$. The **pivots** $1, -5, -0.4$ sit on the diagonal of U .

A preview Strang drops in passing: multiply the pivots and you get the *determinant* of A . Here $1 \times (-5) \times (-0.4) = 2$. Determinants come much later.



The row picture, before and after elimination (use the buttons; drag to rotate). Elimination swaps the return and exposure planes for new ones, ending with the flat plane $z = 2$. All three still meet at the same point (5, 3, 2).

4. Back substitution

$Ux = c$ is triangular, so solve it **from the bottom up**. Each equation has only one new unknown:

- 1 Row 3: $-0.4z = -0.8$, so $z = 2$.
- 2 Row 2: $-5y + 3(2) = -9$, so $-5y = -15$ and $y = 3$.
- 3 Row 1: $x + 3 + 2 = 10$, so $x = 5$.

₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap, the answer Lesson 1 found by substitution. Check it in the **original rules**: $5 + 3 + 2 = 10$ ✓, $60 + 21 + 30 = 111$ ✓, $5 + 0 + 2.4 = 7.4$ ✓.

THE WHOLE METHOD

Forward elimination ($A \rightarrow U, b \rightarrow c$), then **back substitution** ($Ux = c$, bottom row first). That is all a computer does to solve $Ax = b$, just faster and with millions of rows.

▼ **Check yourself (Strang's lecture example):** eliminate $x + 2y + z = 2$, $3x + 8y + z = 12$, $4y + z = 2$.

$\ell_{21} = 3$: row 2 becomes $2y - 2z = 6$. $\ell_{31} = 0$: nothing to do. $\ell_{32} = 4/2 = 2$: row 3 becomes $5z = -10$. Pivots 1, 2, 5. Back substitution: $z = -2$, then $2y + 4 = 6$ so $y = 1$, then $x + 2 - 2 = 2$ so $x = 2$.

5. When elimination breaks down

Pivots can never be 0: we divide by them to find multipliers. So what if a 0 lands in a pivot position? Strang separates two cases.

Temporary failure: fixed by a row exchange

Replace the bond fund with a **credit fund** that also earns 12% but has no market exposure (beta 0). Its column is $(1, 12, 0)$. With target $(10, 126, 7.4)$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 12 & 15 & 126 \\ 1 & 0 & 1.2 & 7.4 \end{array} \right] \xrightarrow{R_2 - 12R_1, R_3 - R_1} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \mathbf{0} & 3 & 6 \\ 0 & -1 & 0.2 & -2.6 \end{array} \right]$$

A 0 in the second pivot position: Nifty and the credit fund earn the same, so removing x from the return rule removed y too. But there is a non-zero entry **below** it, so **exchange rows 2 and 3**:

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-1} & 0.2 & -2.6 \\ 0 & 0 & \boxed{3} & 6 \end{array} \right]$$

Now it is upper triangular, so back-substitute from the bottom row up:

- 1 Row 3: $3z = 6$, so $z = 2$.
- 2 Row 2: $-y + 0.2z = -2.6$. Put in $z = 2$: $-y + 0.4 = -2.6$, so $-y = -3$ and $y = 3$.
- 3 Row 1: $x + y + z = 10$. Put in $y = 3, z = 2$: $x + 5 = 10$, so $x = 5$.

₹5 lakh Nifty, ₹3 lakh credit fund, ₹2 lakh midcap. Check in the original rules: budget $5 + 3 + 2 = 10$ ✓; return $12(5) + 12(3) + 15(2) = 60 + 36 + 30 = 126$ ✓; exposure $5 + 0 + 1.2(2) = 7.4$ ✓.

Three pivots $(1, -1, 3)$ after the exchange, so the matrix is fine (invertible). Swapping the order of two equations never changes the solution.

Permanent failure: the matrix is singular

Now replace the midcap fund with the **balanced fund** from Lesson 1 (half Nifty, half bonds), column (1, 9.5, 0.5):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 9.5 & 111 \\ 1 & 0 & 0.5 & 7.4 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & -2.5 & -9 \\ 0 & -1 & -0.5 & -2.6 \end{array} \right] \xrightarrow{R_3 - 0.2R_2} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & -2.5 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

A 0 in the third pivot position and **nothing below it** to swap in. Only **two pivots**. The last row reads $0 = -0.8$: **no solution**. (With a target on the funds' plane, such as (10, 95, 5), the last row becomes $0 = 0$ and there are infinitely many solutions.)

PIVOTS DECIDE EVERYTHING

An $n \times n$ matrix is **invertible** exactly when elimination (with row exchanges where needed) produces n **non-zero pivots**. Fewer pivots means **singular**: no solution or infinitely many, just as the pictures in Lesson 1 showed.

The 2×2 version: two Nifty funds give $\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, one pivot, and the return rule becomes $0 = -15$.

▼ **Check yourself:** in Strang's example, which number in place of the 8 causes a temporary failure, and which number in place of the last 1 causes a permanent one?

An $8 \rightarrow 6$ makes the second pivot 0, but row 3 has a 4 below it, so a row exchange fixes it. A last entry of -4 instead of 1 makes the third pivot $-4 - (-4) = 0$ with nothing below: permanent failure, singular matrix.

6. Elimination as matrices

Strang's main point: every step of elimination is a **matrix multiplication**. To see why, we need the "row" twin of Lesson 1's key fact.

COLUMNS AND ROWS

Matrix $\times$ column = a combination of the **columns** of the matrix (Lesson 1).

Row $\times$ matrix = a combination of the **rows** of the matrix: $\begin{bmatrix} a & b & c \end{bmatrix} A = a(\text{row 1}) + b(\text{row 2}) + c(\text{row 3})$.

So the row $\begin{bmatrix} -12 & 1 & 0 \end{bmatrix}$ times A gives $-12(\text{row } 1) + 1(\text{row } 2) = (0, -5, 3)$: exactly the new row 2 from step (2, 1).

Elimination matrices

Stack three such rows and you get the matrix that does a whole step. Start from the identity matrix I (which leaves every row alone) and put $-\ell_{ij}$ in position (i, j) :

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.2 & 1 \end{bmatrix}$$

E_{21} subtracts $12 \times$ row 1 from row 2 and leaves rows 1 and 3 unchanged. Check one entry the way Strang does: row 2 of E_{21} times column 3 of A is $-12(1) + 1(15) + 0(1.2) = 3 \checkmark$, the new entry in row 2, column 3.

$$E_{32} (E_{31} (E_{21} A)) = U.$$

Moving the parentheses

Matrix multiplication is **associative**: you may regroup a product without changing it. So the three steps combine into **one matrix that does all of elimination**:

$$(E_{32} E_{31} E_{21}) A = U, \quad E = E_{32} E_{31} E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 1.4 & -0.2 & 1 \end{bmatrix}.$$

Check the bottom row: $1.4(\text{row } 1) - 0.2(\text{row } 2) + 1(\text{row } 3) = (1.4 - 2.4 + 1, 1.4 - 1.4 + 0, 1.4 - 3 + 1.2) = (0, 0, -0.4) \checkmark$, the last row of U .

WATCH OUT: ORDER MATTERS

You may move the parentheses, but you may **not** change the order: in general $AB \neq BA$. For example $E_{21} E_{32}$ has a 0 where $E_{32} E_{21}$ has a non-zero entry. Notice too that E contains a messy 1.4 even though the multipliers were the clean numbers 12, 1, 0.2. Strang's fix, going back from U to A instead, is the subject of Lesson 4 ($A = LU$).

▼ Check yourself: what does the row $\begin{bmatrix} 1 & 2 & 7 \end{bmatrix}$ times a 3×3 matrix produce?

A row: $1 \times \text{row } 1 + 2 \times \text{row } 2 + 7 \times \text{row } 3$ of that matrix. (Strang's example in the lecture.)

7. Row exchanges: permutation matrices

The row exchange in topic 5 is also a matrix. To find it, **do the exchange to the identity matrix**:

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} : \quad P_{23}A \text{ is } A \text{ with rows 2 and 3 swapped.}$$

A matrix that only reorders rows is a permutation matrix. For a 2×2 :

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix} \quad (\text{rows swapped})$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix} \quad (\text{columns swapped})$$

LEFT ACTS ON ROWS, RIGHT ACTS ON COLUMNS

Multiplying **on the left** does row operations (all of elimination). Multiplying **on the right** does column operations. Since PA and AP are different matrices, this is another case of $AB \neq BA$.

In portfolio terms: swapping two **rows** just lists the rules in a different order. Swapping two **columns** relabels which fund is x and which is y .

8. Undoing a step: inverses

E_{21} subtracted $12 \times$ row 1 from row 2. The step that **undoes** it adds $12 \times$ row 1 back:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{21}^{-1}E_{21} = I.$$

The matrix that undoes another is its inverse, written E^{-1} . Every elimination step and every row exchange can be undone like this.

WHY ELIMINATION GIVES THE RIGHT ANSWER

Every step can be reversed, so $Ax = b$ and $Ux = c$ have exactly the same solutions: anything that solves one solves the other. That's why the answer we read off U is the answer to the original portfolio.

Inverses of whole matrices, and how to compute them, are the next lesson.

▼ Check yourself: what is the inverse of E_{32} (which subtracts $0.2 \times$ row 2 from row 3)?

Add $0.2 \times$ row 2 back to row 3: the identity with $+0.2$ in position $(3, 2)$.